

Teaching line, not a question slide.

Plan: opening 0:00, Loop 1 0:04, Loop 2 0:26, Loop 3 0:49, Loop 4 1:12, lose marks and summary 1:42, mini mock 1:44, homework 1:59.

Each loop ends with about 6 minutes of writing on the In-class work sheet (Parts A to D). Part E is the mini mock.

Say it:

reproduction = ree-pruh-DUK-shun

hormones = HOR-mohnz

menstrual = MEN-stroo-ul

fertilisation = fer-tih-ly-ZAY-shun

chromosomes = KROH-muh-sohmz

pregnancy = PREG-nun-see

fertility = fer-TIL-ih-tee

1. sperm (male); egg or ovum (female)
 2. the ovaries
 3. fertilisation
 4. 46 (23 pairs)
 5. the placenta
- Diagnostic:

1 to 3 wrong: go slowly through Loop 1. 4 wrong: spend longer on the chromosome slide in Loop 3. 5 wrong: spend longer on the placenta slides in Loop 4.

Say it:

ovum = OH-vum

ovaries = OH-vuh-reez

placenta = pluh-SEN-tuh

foetus = FEE-tus

Teaching line, not a question slide.

Q fertilisation happens in the oviduct; the zygote starts to divide there on its way to the uterus

Say it:

zygote = ZY-goht

embryo = EM-bree-oh

implantation = im-plan-TAY-shun

oviduct = OH-vih-dukt

uterus = YOO-tuh-rus

1. A tip of the head (it holds enzymes), B nucleus, C middle piece (it holds mitochondria), D tail; P nucleus, Q cytoplasm, R cell membrane, S jelly layer
2. 23 in each
3. the egg is much larger; only the sperm can move: it swims with its tail

Say it:

nucleus = NYOO-klee-us

cytoplasm = SY-toh-plaz-um

membrane = MEM-brayn

enzymes = EN-zymz

mitochondria = my-toh-KON-dree-uh

Teaching line, not a question slide.

Gaps: it swims to the egg; they release energy for swimming; they break through the jelly layer and membrane of the egg; it is a food store for the first cell divisions; it stops any other sperm getting in.

Q very few sperm survive the journey to the oviduct, so millions give a better chance that one meets the egg; one egg is enough because it stays inside the body

Say it:

adaptations = ad-ap-TAY-shunz

1. A ovary, B oviduct (fallopian tube), C uterus, D lining of the uterus (endometrium), E cervix, F vagina
2. in the ovaries
3. in the uterus
4. the cervix

Say it:

ovary = OH-vuh-ree

fallopian = fuh-LOH-pee-un

endometrium = en-doh-MEE-tree-um

cervix = SUR-viks

vagina = vuh-JY-nuh

Teaching line, not a question slide.

Gaps: eggs; fertilisation; foetus; embryo; muscle; sperm (semen).

The two hormones from the ovary are oestrogen and progesterone.

Say it:

oestrogen = EE-struh-jun

progesterone = proh-JES-tuh-rohn

semen = SEE-mun

1. 1 ovary, 2 fallopian tube, 3 bladder, 4 vagina, 5 uterus, 6 rectum, 7 vulva, 8 urethra, 9 anus, 10 cervix

2. bladder and urethra (they store and carry urine); rectum and anus (digestive system)

Say it:

anus = AY-nus

rectum = REK-tum

urethra = yoo-REE-thruh

vulva = VUL-vuh

urine = YOOR-in

1. A bladder, B seminal vesicle, C prostate gland, D sperm duct (vas deferens), E urethra, F epididymis, G testis, H scrotum

2. in the testes

3. the urethra

Say it:

seminal = SEM-ih-nul

vesicle = VES-ih-kul

prostate = PROS-tayt

vas deferens = vas-DEF-uh-renz

epididymis = ep-ih-DID-ih-mis

testis = TES-tis

testes = TES-teez

scrotum = SKROH-tum

Teaching line, not a question slide.

Order: B, D, C, E, A. Gap: semen.

Q sperm are made best at a temperature a little below body

temperature; in the scrotum the testes stay cooler

Say it:

penis = PEE-nis

Teaching line, not a question slide.

Gaps: cooler; sperm, testosterone; stores; urethra; fluid; vagina.

The glands are the seminal vesicles and the prostate gland.

Say it:

testosterone = tes-TOS-tuh-rohn

1. eggs are made in the ovaries, not in the uterus (L4)
2. the foetus develops in the uterus; the cervix is the ring of muscle at its base (L4)
3. sperm are stored in the epididymis; the scrotum is the bag of skin

that holds the testes (L4)

Short solution:

Eggs are made in the ovaries and travel down the oviduct. The uterus is the place where the foetus develops. In a man, sperm are made in the testes and stored in the epididymis.

Q1 any two: a sperm is much smaller; a sperm has a tail and can swim, an egg cannot move; an egg has a large food store in its cytoplasm; millions of sperm are made, one egg is released each month

Q2 sperm are made best at a temperature a little below body temperature; outside the body the testes stay cooler

Q3 fertilisation happens in the oviduct; the sperm cannot reach the egg and the egg cannot reach the uterus, so no zygote forms and nothing can implant

Teaching line, not a question slide.

About 6 minutes, 0:20 to 0:26. The answers are in [inclass_answers.pdf](#).

Teaching line, not a question slide.

Gaps: messengers; blood; testosterone; oestrogen; periods.

Q true

Both sexes: a growth spurt, and pubic and underarm hair.

Say it:

puberty = PYOO-buh-tee

Teaching line, not a question slide.

Gaps: breaks down; period (menstruation); builds up; ovary.

Stage names: 1 menstrual, 2 follicular. In stage 2 a follicle in the ovary grows; FSH (follicle stimulating hormone) makes the egg inside it mature.

Say it:

menstruation = men-stroo-AY-shun

follicular = fuh-LIK-yoo-luh

follicle = FOL-ih-kul

stimulating = STIM-yoo-lay-ting

Teaching line, not a question slide.

Gaps: released; ovulation; thick.

Q around day 14: on the day of ovulation and the two days before and after it

Stage names: 3 ovulation, 4 luteal. With no fertilised egg the lining breaks down and the cycle starts again.

Say it:

ovulation = ov-yoo-LAY-shun

luteal = LOO-tee-ul

Teaching line, not a question slide.

Gaps: mature; pituitary gland, ovulation; builds up; ovary, thick.

Q FSH and LH

Say it:

pituitary = pih-TYOO-ih-tree

luteinising = LOO-tee-in-eye-zing

Teaching line, not a question slide.

1. oestrogen

2. progesterone

3. the lining breaks down and is lost: a period starts

Teaching line, not a question slide.

1. day 14

2. ovulation: an egg is released from an ovary

3. it stays low and almost constant

Teaching line, not a question slide.

Gaps: implants; progesterone; breaks down; vagina.

Q the embryo implants, progesterone stays high and the lining stays thick, so there is no period

1. ovulation is the release of an egg from an ovary; losing the lining is menstruation (L5)
2. the egg is released on about day 14, not day 28 (L5)
3. oestrogen is made in the ovaries, not in the pituitary gland (L6)

Short solution:

Ovulation is the release of an egg from an ovary, on about day 14 of the cycle. Oestrogen is made in the ovaries and makes the lining thicker.

Q1 about days 12 to 16: ovulation is on about day 14, and she is fertile for two days before and after it

Q2 the lining builds up and becomes thicker, with more blood

vessels; oestrogen

Q3 progesterone keeps the lining of the uterus thick, so the lining is not lost; the embryo stays implanted and the placenta can form, so there are no periods during pregnancy

Teaching line, not a question slide.

About 6 minutes, 0:43 to 0:49. The answers are in [inclass_answers.pdf](#).

Teaching line, not a question slide.

Order: B, E, D, A, C.

Q card C is fertilisation; it happens in the oviduct

Say it:

intercourse = IN-ter-kors

ejaculation = ih-jak-yoo-LAY-shun

Teaching line, not a question slide.

Gaps: zygote; embryo; implantation; foetus.

Q the membrane of the egg changes as soon as one sperm has entered

Teaching line, not a question slide.

Gaps: sperm 23; egg 23; zygote 46; every body cell 46 (23 pairs); nucleus.

Q so that the zygote has the full 46 and not 92 when the two sex cells join; half comes from each parent

Say it:

genes = JEENZ

Teaching line, not a question slide.

Gaps: X; X or Y.

1. top row XX, XX; bottom row XY, XY

2. 50 % (2 of the 4 boxes)

3. the father's: a sperm with X gives a girl, a sperm with Y gives a boy

Teaching line, not a question slide.

1. row X
2. one, one; splits into two
3. two, two

4. row Y, the non-identical (fraternal) twins: they have different genes; identical twins have the same genes, so they are always the same sex

Say it:

identical = eye-DEN-tih-kul

fraternal = fruh-TUR-nul

- (a) true
- (b) false: they are always the same sex
- (c) true

- (d) true
- (e) false: that gives non-identical twins
- (f) false: each egg must be fertilised by a sperm

Q1 in the oviduct (fallopian tube)

Q2 they die and are broken down

Q3 on land the sex cells would dry out; inside the body the sperm

can swim in fluid, the sex cells are protected, and a sperm is much more likely to meet the egg

Say it:

evolutionary = ee-vuh-LOO-shun-ree

Gaps: X; Y; 50 % (1 in 2); chance.

Each fertilisation is a new event with a new sperm cell.

Teaching line, not a question slide.

1. a sperm and an egg have 23 chromosomes each, so the zygote has 46 (L1)
2. the sperm decides the sex: it carries X or Y, and every egg carries X (L7)
3. identical twins come from one fertilised egg that splits; two eggs

give non-identical twins (L11)

Short solution:

A sperm has 23 chromosomes and an egg has 23, so the zygote has 46. The father's sperm decides if the baby is a boy or a girl. Identical twins form when one fertilised egg splits into two.

Q1 in the oviduct; a zygote

Q2 no: identical twins come from one zygote, so they have the same chromosomes and are the same sex; a boy and a girl must

come from two eggs and two sperm

Q3 the child gets 23 chromosomes from the mother's egg and 23 from the father's sperm, so its genes are a mixture of both; the mixture is a new combination, different from each parent

Teaching line, not a question slide.

About 6 minutes, 1:06 to 1:12. The answers are in [inclass_answers.pdf](#).

1. A placenta, B umbilical cord, C amniotic fluid, D amniotic sac, E uterus wall, F cervix

2. the amniotic fluid

3. the umbilical cord

Say it:

umbilical = um-BIL-ih-kul

amniotic = am-nee-OT-ik

Teaching line, not a question slide.

Down: oxygen; glucose (food, nutrients); antibodies. Up: carbon dioxide; urea.

Gaps: mix; diffusion.

Say it:

villi = VIL-eye

glucose = GLOO-kohz

antibodies = AN-tee-bod-eez

urea = yoo-REE-uh

diffusion = dih-FYOO-zhun

- (a) the mother and the foetus can have different blood types; the mother's blood pressure is higher and would damage the foetus's small blood vessels; keeping the bloods apart stops many bacteria and harmful substances reaching the foetus
- (b) for respiration, which releases the energy its cells need to divide and grow
- (c) carbon dioxide and urea; they are waste products and would

poison the foetus if they built up

Short solution:

Blanks: exchange (diffusion); mixing.

Say it:

foetal = FEE-tul

maternal = muh-TUR-nul

respiration = res-pih-RAY-shun

Teaching line, not a question slide.

Gaps: higher; diffuses; surface area; vein.

Q the foetus's blood, arriving in the umbilical artery, has a higher concentration of carbon dioxide than the mother's blood, so carbon

dioxide diffuses across the villi into the mother's blood, which carries it away

Say it:

concentration = kon-sun-TRAY-shun

artery = AR-tuh-ree

Teaching line, not a question slide.

Gaps: oxygen; brain.

Q the foetus is very small and is growing very quickly

Cigarette smoke: carbon monoxide takes the place of oxygen in the mother's blood; nicotine also crosses the placenta.

Say it:

rubella = roo-BEL-uh

addicted = uh-DIK-tid

monoxide = mon-OK-syd

nicotine = NIK-uh-teen

Teaching line, not a question slide.

Gaps: 40; 8 (the glossary says after 8 weeks, page 28 says nine);
20; 38.

Q premature

After 20 weeks the loss of a baby is called a stillbirth.

Say it:

gestation = jes-TAY-shun

miscarriage = MIS-ka-rij

premature = PREM-uh-chur

Teaching line, not a question slide.

Order: B, C, A. Stage 1 labour; stage 2 delivery; stage 3 afterbirth.

Q the amniotic fluid

Say it:

dilates = dy-LAYTS

labour = LAY-bur

1. doctors try to widen the oviducts; or IVF
2. doctors try to widen the sperm ducts
3. IVF
4. IVF

5. inject a sperm cell into an egg cell with a fine needle

Short solution:

These are the answers in the paragraphs on page 31. The numbers 1 to 5 are the five empty rows, from the top.

Teaching line, not a question slide.

Order: C, A, D, E, B.

Q often more than one embryo is put into the uterus, and each one

can implant

In vitro means in glass: fertilisation happens outside the body.

Say it:

vitro = VEE-troh

Q3 (a) from 1960 to about 1990 it stayed between about 11 % and 15 %; after 1990 it rose steadily to about 25 % in 2010
Q3 (c) it stayed almost the same, at about 9 % to 10 %, from 1960 to

2010

Short solution:

A multiple birth is two or more babies born from one pregnancy.

Q3 (b) fertility treatment has become more common; older women are more likely to need it; fertility treatment increases the chance of a multiple birth

Q3 (d) women aged 20 to 24 rarely have problems becoming pregnant, so few of them have fertility treatment

Q4 (a) about 30 % or a little more: any value above 25 % that continues the rising line

Q4 (b) I extended the trend line from 1990 to 2010 forward to last year

Say it:

hypothesis = hy-POTH-uh-sis

1. the two bloods never mix: only the foetus's blood flows in the umbilical cord (L9)
2. diffusion is from a high concentration to a low concentration (L10)
3. the amniotic fluid does not feed the foetus: food comes from the mother's blood through the placenta (L13)

Short solution:

Oxygen diffuses from the mother's blood into the foetus's blood in the placenta, from a high concentration to a low concentration. The foetus's blood carries it along the umbilical cord. The amniotic fluid protects the foetus from bumps.

Q1 carbon dioxide and urea

Q2 carbon monoxide from the smoke is carried in the mother's blood, so less oxygen crosses the placenta; the foetus's cells

respire less and release less energy, so it grows more slowly

Q3 a multiple birth is riskier for the mother and the babies, who are often premature and small; with one embryo only one can implant, so there are fewer twins and triplets

Teaching line, not a question slide.

About 6 minutes, 1:36 to 1:42. The answers are in [inclass_answers.pdf](#).

Teaching line, not a question slide.

1. 23: every sex cell has one set; the zygote and body cells have 46 (L1)
2. in the oviduct; the embryo implants in the uterus later (L2)
3. ovulation is the release of an egg; losing the lining is

menstruation (L5)

4. the bloods never mix; oxygen and food diffuse across the placenta (L9)
5. add how it helps: so that it can swim to the egg (L3)
6. describe needs the trend and numbers from the graph; the reason is for suggest or explain (L12)

Teaching line, not a question slide.

Ask her to say each line with the book closed, then give one example question for each.

Teaching line, not a question slide.

1:44 to 1:56 writing, 1:56 to 1:59 marking. The answers are in [inclass_answers.pdf](#).

The answers are in [homework_answers.pdf](#).

Teaching line, not a question slide.